

Regular Progress Report: Cloud-Based Movie Streaming Platform

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I. SUMMARY OF THIS WEEK

This week focused on two user-facing improvements:

- 1) Dynamic, relevance-based search (not exact text-only matching).
- 2) Auto-hide mouse cursor behavior in the video player, synchronized with player controls.

II. FEATURE 1: DYNAMIC SEARCH UPGRADE

A. Problem

The previous search system only returned results when the query closely matched movie titles. This created poor UX for typo inputs and near matches.

B. Work Completed

- Reworked backend movie search endpoint to use ranking/scoring logic instead of strict text filtering.
- Added normalization (lowercasing and punctuation cleanup).
- Added fuzzy similarity scoring using string similarity.
- Added weighted ranking signals:
 - Exact title match
 - Prefix/contains match
 - Token overlap
 - Similarity on title and description
- Preserved pagination and existing frontend API contract.

C. Result

Search now returns relevant alternatives even when the exact movie title is not typed (e.g., typo-tolerant behavior similar to commercial streaming UX).

III. FEATURE 2: PLAYER CURSOR AUTO-HIDE

A. Problem

During playback, the cursor remained visible even when controls were hidden, reducing immersion.

B. Work Completed

- Updated player UI state logic so cursor visibility follows control visibility.
- Cursor now hides during active playback after inactivity timeout.
- Cursor reappears immediately on user activity (mouse movement/touch), together with controls.
- Behavior is restricted to player interaction state and does not affect other pages.

C. Result

Playback experience is now cleaner and immersive.

IV. VALIDATION

- Verified dynamic search with typo queries and partial matches.
- Verified cursor hide/show transitions across play, pause, and activity states.
- Confirmed no API contract break for frontend integration.

V. NEXT STEPS

- 1) Fine-tune search ranking weights based on real usage.
- 2) Add optional search analytics (top queries, zero-result terms).
- 3) Add keyboard-accessibility refinements for player controls.